Three Bounds on a Null: Testing the Link Between Fast-Weight Write Geometry and In-Context Composition

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Abstract

Fixed-state sequence models bind in-context associations in a matrix-valued fast-weight state, and geometric observables of that state, such as the span occupied by key writes, have been proposed as lenses on composition. We test whether one such observable predicts, or causally improves, in-context multi-hop composition in DeltaNet-family language models spanning 14M to 1.31B parameters, and we find a null bounded three ways. First, a state-space composition readout returns a recovered fraction of exactly zero at 366 of 366 readings across three structurally different instruments (zero-shot marker prompts, natural language, and task-familiarized checkpoints); the readout's pre-registered validity gates fail at every cell, so the zeros are an instrument verdict, not a refutation. Second, a vocabulary-space behavioral contrast bounds any causal effect of a write-geometry intervention below its pre-registered three-seed detection floor of 1.5 to 1.7 loss units; the sole reading meeting the pre-registered differential condition (one of five that cross zero across forty tests) is a transient, harm-direction deviation. Third, a twelve-seed replication of the transient was stopped by its pre-registered batch-effect gate (cohort variance ratio 4.47 against a heuristic cutoff of 4.0), and the new cohort's interval spans zero. Each bound names the observation that would overturn it. The result is a null with its assumptions visible and its escape routes closed.

1 Introduction

Linear-attention and delta-rule language models compress an entire context into a fixed-size matrix state (Schlag et al., 2021; Yang et al., 2024). A companion research program measured geometric observables of the written state (the span fraction occupied by key writes, the effective rank of the key population) across scales and under targeted interventions, carrying an unexamined promise: that write geometry matters for something a user would recognize, such as composing multi-hop relations presented in context. We test that promise and find a null. A reader of a null has two standing worries, that the instrument never worked and that the analysis could declare a null after seeing the data. Three pre-registered bounds address both:

1. An instrument bound. A state-space composition readout, deployed three structurally different ways, returns a recovered fraction of exactly zero at 366 of 366 readings on checkpoints from 14M to 1.31B parameters, and its pre-registered validity gates fail at every cell, routing the outcome to probe-invalid, not refutation. One nonzero reading or one gate pass would have voided this bound.
2. A power bound. A vocabulary-space behavioral contrast, with a three-seed detection floor of 1.5 to 1.7 loss units registered before unblinding, leaves three of four contrasts unresolved; the sole reading meeting the pre-registered differential condition is a transient, harm-direction deviation. A determinate effect above the floor would have voided this bound.
3. A replication bound. A targeted twelve-seed extension of that transient was stopped by its own pre-registered batch-effect gate (a 4.47-fold cohort variance mismatch against a

Wave (instrument)	Grid	Readings	Registered outcome
1 zero-shot, marker	78 cells, 14M–392M	$312/312 = 0.0$	probe-invalid
2 zero-shot, 1.31B rung	2 cells	$8/8 = 0.0$	probe-invalid
3 zero-shot, natural lang.	4 gate cells	$16/16 = 0.0$	probe-invalid, refused
4 familiarized gate	6 cells × 5 ckpts	$30/30 = 0.0$; L_q falls	dissociation
5 behavioral contrast	18 cells, 4 contrasts†	3 unresolved, 1 transient	power-bounded
6 seed extension, n=12	18 pretrain cells	variance ratio 4.47	batch-effect-flagged

Table 1: The program at a glance. Waves 1–4 deploy the geometric readout (recovered fraction at cosine ≥ 0.9); its validity gates fail at every cell of every wave, so all zero readings route to probe-invalid rather than refutation. Waves 5–6 deploy the vocabulary-space contrast; the one determinate transient is in the harm direction and its n=12 replication was stopped by the pre-registered batch-effect gate. †Per-arm re-derivation; the registered corpus-level pipeline returned both corpora unresolved (Appendix A).

heuristic 4.0 cutoff), and the new cohort's interval spans zero. A clean pooled interval excluding zero would have voided this bound.

The contribution is the bounded null and the measurement discipline that kept it interpretable.

2 Models, Task, and Instruments

Checkpoints and task. All experiments probe DeltaNet-family language models (Yang et al., 2024) pretrained on a reasoning-dense corpus mix (OpenR1-Math) and an encyclopedic mix (WikiText-103, Merity et al., 2017). An intervention grid (14M parameters) varies a frozen-key-bias intervention on the fast-weight write path in three validation-loss-matched arms: control, per-token, and global. A scale ladder spans 14M, 98M, 392M, and 1.31B parameters (state dimension 64, 64, 128, 128; depth 2, 12, 16, 22). Each probe episode binds K entities into one directed cycle with bind statements, then queries the entity $h \in \{1, \ldots, 4\}$ hops around the cycle; a full K-cycle prevents held-out hop depths from collapsing modulo short cycle lengths. Load is K relative to state dimension ($K \in \{20, 32\}$ on the grid; near-capacity per rung).

Instrument 1: a geometric readout. For a layer's terminal state S_T and effective query q_{eff}, the readout predicts the h-hop target as $S_T^h q_{\text{eff}}$ and scores exact continuous recovery: the fraction of queries reaching cosine at least 0.9 with the target value vector. Continuous recovery, rather than argmax over candidates, follows Nichani et al. (2024), under which argmax decoding can succeed without the state carrying the claimed structure. Registered validity gates accompany the readout: an h=1 sanity floor and two alignment premises, (iii) query–key and (iv) key–value, each requiring a same-entity median cosine above a shuffled-entity null's 95th percentile (Appendix A); gate failure routes to probe-invalid, never to refutation.

Instrument 2: a behavioral contrast. A familiarization phase trains checkpoints for 5,000 steps on the bind/query task mixed into the pretraining corpus; the contrast is the held-out-episode query loss L_q (a vocabulary-space cross-entropy through the language-model head, machinery disjoint from Instrument 1), differenced control minus arm at five checkpoints. A six-outcome trajectory classification and a differential condition (the K=32 effect determinate while K=20 is not) were fixed before unblinding. Every wave ran gate-first, with verdicts routed mechanically (Appendix A).

3 The Geometric Readout Never Fires

Zero-shot, marker template (wave 1). The full pre-registered grid, 60 intervention cells and 18 ladder cells (Table 1), returns a recovered fraction of exactly 0.0 at all 312 (cell, hop) readings. The validity gates fail everywhere: the h=1 floor fails both conditions, and premises (iii) and (iv) pass at 0 of 78 cells each. Premise medians track their shuffled nulls but never cross them (premise (iii) spans $[-0.33, 0.76]$), and prediction–target cosines center near zero, ten-plus standard deviations from the 0.9 threshold (Figure 1, Appendix B).

Scale (wave 2). The 1.31B rung reproduces the failure in kind: 8 of 8 readings at 0.0, both premise gates failing at both corpora, for a ladder series of 80 of 80 zeros across 14M to 1.31B: not a small-model artifact.

Zero-shot, natural language (wave 3). Two template families, a gift-verb family with the out-of-distribution query marker dropped and a corpus-native succession-verb family, fail identically: 16 of 16 readings at 0.0, the $h=1$ gate false at all 4 cells. The failure is not attributable to prompt surface form.

Task-familiarized (wave 4). The strongest-case test trains control checkpoints on the task itself for 5,000 steps and applies the gate at five checkpoints per cell. It fails at 30 of 30 readings, 0 of 512 queries recovered each time, while the vocabulary-space query loss falls 21.8 to 46.4 percent (mean 35.9 percent) in 6 of 6 cells, measured from checkpoint 250, the first post-warmup reading, to checkpoint 5,000 (Figure 2, Appendix B): the models measurably learn the task through one readout while the other stays at floor. Since the two share no machinery by construction, the dissociation indicts the readout construct, not the models (the fall stops short of a registered 50-percent pin, so the adjudication claims partial task learning only).

Gates with enforcing code paths. Wave 1's launch chain computed its abort gate but had no code path acting on it; the grid ran past a gate that had already declared the probe invalid. Every later chain enforced its gates at the process boundary; waves 3 and 4 each refused a full-grid launch on a real failing gate.

4 The Behavioral Contrast Is Power-Bounded

Wave 5 promotes the vocabulary-space contrast to the primary instrument: all three arms are familiarized under identical recipes (18 cells), and the arm effect is the held-out query-loss difference, control minus arm, so positive means the intervention helps.

The registered power floor. Before unblinding, the three-seed detection floor was derived from the control cells' between-seed spread: a σ proxy of 0.43 to 0.48 loss units gives a 95-percent half-width of roughly 1.5 to 1.7. For calibration, the entire familiarization effect is about 1.69 loss units: the instrument detects only effects about as large as everything the model learns in the phase, so the registration anticipated unresolved outcomes, to be read as measurement bounds, never as evidence of no effect.

Results. The registered corpus-level classifier, which uses the global arm as each corpus's representative, returns both corpora unresolved; a disclosed build-time per-arm re-derivation (folded into the analysis code, re-validated against the archived deltas; Appendix A) splits this into four contrasts, three unresolved. The fourth (encyclopedic corpus, per-token arm) classifies as transient: at checkpoint 2,500 the $K=32$ effect is determinate at -0.500, interval $[-0.624, -0.376]$, while $K=20$ is not, and by checkpoint 5,000 it dissolves (Figure 3, left, Appendix B). A held-out-hop secondary readout fires at the same cell and checkpoint, same direction, consistent with one coherent mid-training deviation.

Sign discipline and multiplicity. Of 40 uncorrected 95-percent intervals in the primary table, five exclude zero against two expected by chance, all in the harm direction (arm loss above control); four of the five sit at checkpoint 1,000, both loads of two different cells, consistent with checkpoint-level sampling noise (Appendix A). Only the fifth, at checkpoint 2,500, also satisfies the pre-registered differential condition and draws the held-out-hop corroboration; that compound condition, not any single interval, is the one determinate signal. "The intervention matters" ignores that transients were registered as training-dynamics artifacts; "no effect" ignores the power floor.

5 The Replication Gate Refuses the Pool

Wave 6 extends the transient's anchor cell (encyclopedic corpus, per-token arm, $K=32$, checkpoint 2,500) to twelve seeds with three registered integrity mechanisms: nine fresh pretraining seeds per arm (18 cells), an archived-values loader whose runtime guard forbids live re-scoring, and a pre-pooling batch-effect gate (means within twice the pooled standard error; variance ratio below 4.0).

The gate fired. The archived cohort's between-seed standard deviation (1.154) is 2.1 times the new cohort's (0.546): variance ratio 4.47, past the heuristic 4.0 cutoff (a routing threshold, not a calibrated test; the one-sided 5-percent $F(2, 8)$ value is ≈ 4.46); the means agree (shift 0.364, threshold 1.382). The registered routing reports the cohorts separately, batch-effect-flagged, with no decision-grade pooled interval: the archived cohort reproduces its original $[-0.624, -0.376]$ by construction; the new nine-seed cohort's own interval, $[-0.506, +0.357]$, spans zero; and the naive twelve-seed pool, diagnostic only, reads $[-0.509, +0.147]$, a refutation absent the gate (Figure 3, right, Appendix B).

What stands. The transient is neither confirmed nor refuted; it keeps its three-seed weight. Every computable pooled interval, at 7 of 10 primary and 7 of 10 held-out (checkpoint, load) pairs, contains zero. The variance mismatch is not established as systematic; its direction flips across checkpoints, consistent with small-n imprecision in the archived cohort.

6 Related Work

Zoology (Arora et al., 2023) and Based (Arora et al., 2024) establish the recall–state-size tradeoff and the ladder pattern this program follows; both train architectures for the probe setting, where this program probes general-purpose checkpoints post hoc and reports instrument validity, not a capacity frontier. Okpekpe & Orvieto (2025) show that optimizer choice, not architecture alone, drives much of the reported Transformer–recurrent recall gap; this paper bounds a different axis, write-geometry validity, and makes no positive mechanism claim. Grazzi et al. (2024) prove positive-eigenvalue DeltaNet cannot compose certain groups (e.g. parity), a ceiling lifted only by negative-eigenvalue transitions; Merrill et al. (2024) give the analogous TC^0 ceiling for diagonal SSMs. Nichani et al. (2024) supply the associative-memory analysis behind the exact-recovery readout (Section 2). Reporting follows the pre-registration workshop (Bertinetto et al., 2021) and negative-results tradition (Forde et al., 2020).

7 Scope of the Bounds

Bounded. The readout construct, one layer's terminal state applied h times to a query under exact continuous recovery, carries no signal in this checkpoint family at any tested scale, surface form, or training regime. Intervention effects beyond 1.5 to 1.7 loss units either way are excluded at the registered confidence; the transient is bounded in duration, direction, and weight.

Not bounded. The hypothesis survives below the power floor and through mechanisms the readout cannot see, most plausibly cross-layer hand-off (one layer resolves the first hop, a later layer the next); the readout tests only single-layer self-iteration. A competing account is expressivity: Grazzi et al. (2024) prove positive-eigenvalue DeltaNet cannot compose certain permutation groups, so the null may partly reflect this variant's ceiling rather than a wrong observable; wave 4's dissociation points toward the observable; and nothing here transfers beyond the DeltaNet family or the bind/query grammar. Finally, the readout's recovery math is validated only at unit level, never end-to-end on a real forward pass with a known target, so a systematic extraction or state-convention error is not fully excluded (Appendix A); a real-kernel positive control is the next wave's priority.

Lessons for small-scale practice. A registered gate without an enforcing code path is a sentence, not a safeguard (refusal worked twice on real failures); routing fixed before unblinding kept zeros from becoming a refutation and a transient from becoming a headline.

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A Registered Gates and Routing Rules

This appendix reproduces, in compact form, the pre-registered gates and mechanical routing rules the main text relies on. Each was fixed in the program's design registry before the corresponding data existed.

Geometric-readout validity gates (waves 1–4). A cell's readout is valid only if all of the following hold. (a) h=1 sanity floor: the one-hop recovered fraction must exceed the upper edge of a shuffled-pairing null by more than the null's width, and must also exceed an absolute floor of 0.10. (b) Premise (iii): the median cosine between an entity's query projection and the same entity's key projection must exceed the 95th percentile of a shuffled-entity null. (c) Premise (iv): the median cosine between the key and value projections of the same token must exceed the same null percentile. Routing: failure of any gate routes the cell to probe-invalid; the registered rule states that gate failure never routes to refutation. The readout's recovery math (state-power application and the cosine-recovery scorer) is unit-tested against hand-built matrices; it has no positive control run end-to-end through a real forward pass with a known target on the production multi-layer kernel path. The failure's uniformity across four structurally different instruments and four scales is consistent with a genuine construct-validity gap but does not, on its own, rule out a systematic extraction or state-convention error producing the same uniform pattern.

Behavioral-contrast classification (wave 5). The per-checkpoint arm effect is $\Delta(c) = L_q(\text{control}, c) - L_q(\text{arm}, c)$, with three-seed 95-percent intervals ($t(2, .975) = 4.303$). A checkpoint satisfies the differential condition when the K=32 interval excludes zero, the K=20 interval does not, and $|\Delta_{32}| > |\Delta_{20}|$. Trajectories classify into six registered outcomes: persistent, transient (the condition holds at an interior checkpoint and fails at the terminal one), late-emergent, converged-equivalent, unresolved, and a familiarization-null exclusion. The three-seed power floor (a detectable effect of roughly 1.5 to 1.7 loss units, derived from the control cells' between-seed spread) was registered together with the instruction that unresolved outcomes be reported as measurement bounds.

The registered pipeline computed one classification per corpus, using the global arm as that corpus's representative signal, and classified both corpora unresolved. The four-way (corpus, arm) breakdown reported in the main text and Table 1 applies the same registered classification primitives to each arm's own trajectory. This finer split was a build-time re-derivation performed after the corpus-level output was seen; it was later folded into the analysis code and re-validated against the archived per-cell deltas, reproducing the same four verdicts. It is disclosed here as a re-derivation, not presented as the registered corpus-level verdict.

The primary table computes 40 uncorrected 95-percent intervals (4 contrasts × 5 checkpoints × 2 loads); five exclude zero: both loads of reasoning-dense×global at checkpoint 1,000 (K=32: Δ=-0.203, $[-0.402, -0.004]$; K=20: Δ=-0.199, $[-0.325, -0.074]$), both loads of encyclopedic×per-token at checkpoint 1,000 (K=32: Δ=-0.168, $[-0.301, -0.036]$; K=20: Δ=-0.109, $[-0.211, -0.006]$), and the K=32 interval of encyclopedic×per-token at checkpoint 2,500 (Δ=-0.500, $[-0.624, -0.376]$, the reported transient). Only the last also satisfies the compound differential condition above and is corroborated by the independent held-out-hop readout at the same cell and checkpoint; that compound condition, not any single interval exclusion, is what the main text calls the one determinate signal. The checkpoint-1,000 cells fail the differential condition precisely because their K=20 intervals also exclude zero.

Replication integrity gates (wave 6). Archived seeds load from stored values only; a runtime guard raises on any live re-scoring of an archived seed. Before pooling, a batch-effect gate requires (a) cohort mean shift below twice the pooled standard error (Welch form) and (b) cohort variance ratio below a pre-registered heuristic 4.0 (not a calibrated F-test; the one-sided 5-percent $F(2, 8)$ critical value for the archived three-seed versus new nine-seed cohorts is ≈ 4.46). On gate failure the registered routing reports both cohorts separately, produces no decision-grade pooled interval, and classifies the wave as batch-effect-flagged, outside the four-outcome partition (confirmed at magnitude, confirmed smaller, refuted, new pattern) that a passing gate would have entered.

B Supplementary Figures

C Reproducibility

Every number in this paper is computed from archived raw result files (per-cell JSON records written by the experiment chains at run time), not from summaries. All figures regenerate from one versioned script that loads only those archived files and asserts the md5 checksum of every file it reads against a recorded manifest, so a changed or substituted artifact fails the build rather than silently altering a panel. As a compute disclosure for this venue's small-scale remit: all experiments ran on one or two GPUs at a time, and the six waves consumed approximately 9.5 GPU-hours in total. The archived raw files, the figure script with its checksum manifest, the experiment chains, and the full pre-registration text will be released with the camera-ready version.

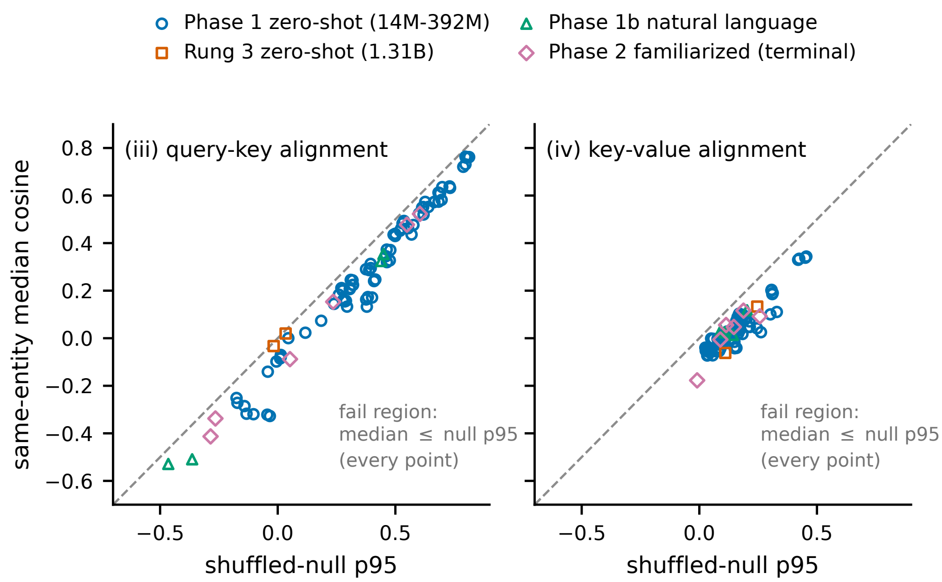

Figure 1: Validity-gate failure at every cell of every instrument. Each point is one cell's same-entity median cosine for premise (iii) (query–key, left) or premise (iv) (key–value, right) against that cell's shuffled-entity null p95; passing requires a median above the dashed identity line. All 90 cells across the four geometric-readout waves fall on or below the line, at every scale from 14M to 1.31B and under every surface form and training regime.

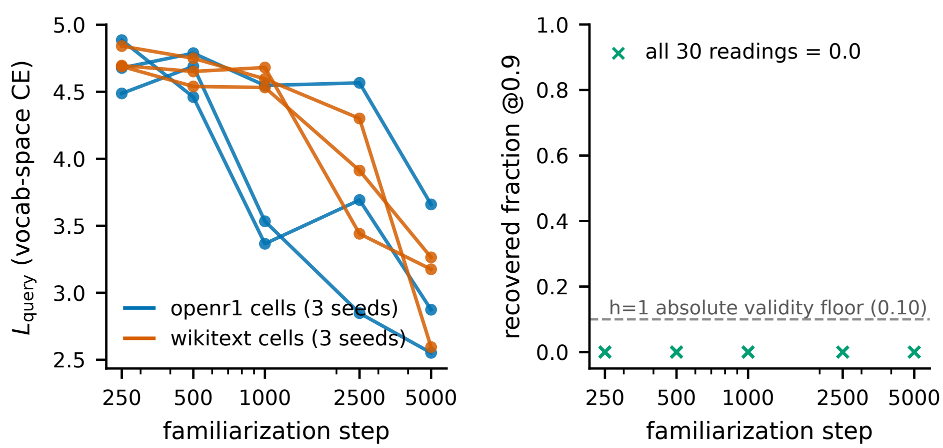

Figure 2: Vocabulary/geometry dissociation under task familiarization (wave 4). Left: held-out in-context query loss L_q for all six control-arm cells (two corpora, three seeds) falls by 21.8 to 46.4 percent between checkpoint 250 (the first post-warmup reading) and checkpoint 5,000. Right: the geometric gate's recovered fraction at the same five checkpoints of the same cells is exactly 0.0 at all 30 readings, never approaching its 0.10 validity floor (dashed). The model learns the task through one readout while the other stays at floor.

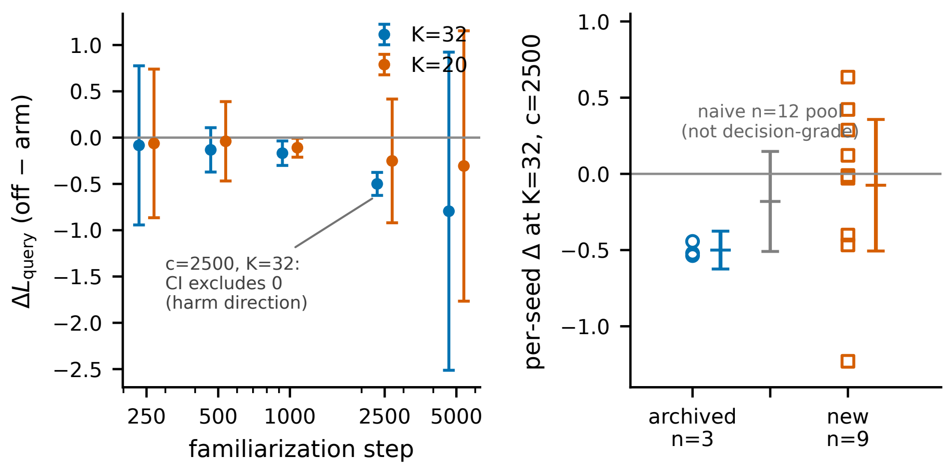

Figure 3: The transient and its stopped replication. Left: per-checkpoint arm effect ΔL_q (control minus per-token arm, encyclopedic corpus) with three-seed 95-percent intervals at both loads; the single reading satisfying the registered differential condition is at checkpoint 2,500, K=32 (-0.500, $[-0.624, -0.376]$), in the harm direction, against a non-determinate K=20 reading at the same checkpoint (mean -0.252, $[-0.920, +0.416]$), and dissolves by checkpoint 5,000 (mean -0.795, $[-2.513, +0.923]$). Right: the same anchor cell at n=12; the archived three-seed cohort (circles, interval $[-0.624, -0.376]$) and the new nine-seed cohort (squares, interval $[-0.506, +0.357]$) fail the pre-registered variance-ratio gate (4.47 against a pre-registered heuristic cutoff of 4.0; a calibrated one-sided 5-percent $F(2, 8)$ test would itself require ≈ 4.46), so the naive pooled interval (gray, dotted) is diagnostic only and no decision-grade pooled verdict exists.